

## **INTRODUCTION TO CLOUD COMPUTING**

Cloud computing has transformed how organizations build, deploy, and scale technology. Instead of owning physical infrastructure, users access computing resources over the internet on demand.

- Eliminates the need for physical servers and upfront hardware costs.
- Enables on-demand access to computing power and storage.
- Supports scalability, flexibility, and cost efficiency.

### **Problem That Cloud Computing Solves**

Before the cloud, if a company wanted to launch a website or an application, it had to follow a slow and expensive process:

#### **The "Old Way" (On-Premises)**

- Companies had to purchase physical servers, storage devices, and networking equipment.
- This required a large upfront investment known as Capital Expenditure (CapEx).
- Peak usage had to be estimated in advance, often leading to overprovisioning and wasted resources.
- Procuring and setting up new servers could take weeks or even months.

#### **The "New Way" (Cloud Computing)**

- Hardware is no longer purchased; computing resources are rented from cloud providers such as AWS, Google, or Microsoft.
- Costs shift from a large upfront investment to a pay as you go Operational Expenditure model.
- Resources can be provisioned instantly without long waiting periods.
- Eliminates overprovisioning, guesswork, and resource wastage.

### **Cloud Computing Architecture**

[Cloud computing architecture](#) refers to the components and sub-components required for cloud computing. These components typically refer to:

1. Front end ( Fat client, Thin client)
2. Back-end platforms ( Servers, Storage )
3. Cloud-based delivery and a network ( Internet, Intranet, Intercloud )

## Cloud Computing Architecture

### **1. Front End ( User Interaction Enhancement )**

The User Interface of Cloud Computing consists of 2 sections of clients. The Thin clients are the ones that use web browsers facilitating portable and lightweight accessibilities and others are known as Fat Clients that use many functionalities for offering a strong user experience.

### **2. Back-end Platforms ( Cloud Computing Engine )**

The core of cloud computing is made at back-end platforms with several servers for storage and processing computing. Management of Applications logic is managed through servers and effective data handling is provided by storage. The combination of these platforms at the backend offers the processing power, and capacity to manage and store data behind the cloud.

### **3. Cloud-Based Delivery and Network**

On-demand access to the computer and resources is provided over the Internet, Intranet, and Intercloud. The Internet comes with global accessibility, the [Intranet](#) helps in internal communications of the services within the organization and the [Intercloud](#) enables interoperability across various cloud services. This dynamic network connectivity ensures an essential component of cloud computing architecture on guaranteeing easy access and data transfer.

## **Characteristics of Cloud Computing**

All cloud services share five fundamental characteristics that define them:

1. **On-Demand Self-Service:** Users can provision computing resources like servers and storage automatically, without requiring human intervention from the service provider.
2. **Broad Network Access:** Capabilities are available over the network and can be accessed through standard mechanisms by any device, such as laptops, tablets, and mobile phones.
3. **Resource Pooling:** The provider's computing resources are pooled to serve multiple customers using a multi-tenant model. Resources are dynamically assigned and reassigned according to demand.
4. **Rapid Elasticity (Scalability):** Resources can be scaled up or down quickly and, in some cases, automatically, to meet demand. This ensures you have enough power during traffic spikes and aren't paying for idle resources during quiet periods.
5. **Measured Service (Pay-as-you-go Model):** Cloud systems automatically control and optimize resource use. Usage is monitored, controlled, and reported, providing transparency for both the provider and the consumer. You only pay for what you use.

### The Shared Responsibility Model

Security in the cloud is a partnership between the cloud provider and the customer. This is known as the Shared Responsibility Model.

- **The Cloud Provider (e.g., AWS, Azure, GCP)** is responsible for the security *OF* the cloud. This includes the physical security of the data centers, the hardware, and the core networking infrastructure.
- **The Customer (You)** is responsible for security *IN* the cloud. This includes managing who has access to your resources, encrypting your data, configuring firewalls (like Network Policies), and securing your applications.

The amount of responsibility you have depends on the service model you choose.